

## Entry: 1

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

5a23 = fault • 14ab = fault • 257e = fault • 7988 = 0x3a8 • 75cf = 0x02f • 3350 = 0x810 • 0a70 = 0x0f0 • 55f9 = 0x0d9

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

30.0

#### Score 3:

30

#### Score outlier:

13

## Entry: 2

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y  
Virtual Address 685e: Translates to what Physical Address (and fetches what value)? Or Fault? Virtual Address 685e: --&

#### Answer:

Virtual Address 5a23 ---- 0x0823

■Virtual Address 14ab ---- Fault

■Virtual Address 257e ---- 0x0c7e

■Virtual Address 7988 ---- 0x7f88

■Virtual Address 75cf ---- 0x7fcf

■Virtual Address 3350 ---- Fault

■Virtual Address 0a70 ---- Fault

■Virtual Address 55f9 ---- Fault

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

0.0

#### Score 3:

8

## Entry: 3

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: --&gt; pde index:0x17 [decimal 23] pde contents:0x7f (valid 0, pfn 0x7f [decimal 127]) --&gt; Fault (p

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

25.0

#### Score 3:

20

## Entry: 4

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run:

```
$ ./paging-multilevel-translate.py -s 3103
```

For each of the following virtual addresses, write down the physical address it translates to or write down that it is a fault (

Virtual Address 5a23

Virtual Address 14ab

Virtual Address 257e

Virtual Address 7988

Virtual Address 75cf

Virtual Address 3350

Virtual Address 0a70

Virtual Address 55f9

#### Answer:

Virtual Address 5a23:

--&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53])

--&gt; pte index:0x11 [decimal 17] pte contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

Virtual Address 14ab:

--&gt; pde index:0x5 [decimal 5] pde contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

Virtual Address 257e:

--&gt; pde index:0x9 [decimal 9] pde contents:0xc4 (valid 1, pfn 0x44 [decimal 68])

--&gt; pte index:0xb [decimal 11] pte contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

Virtual Address 7988:

--&gt; pde index:0x1e [decimal 30] pde contents:0xbb (valid 1, pfn 0x3b [decimal 59])

--&gt; pte index:0x8 [decimal 8] pte contents:0x19 (valid 0, pfn 0x19 [decimal 25])

--&gt; Fault (page table entry not valid)

Virtual Address 75cf:

--&gt; pde index:0x1d [decimal 29] pde contents:0xc6 (valid 1, pfn 0x46 [decimal 70])

--&gt; pte index:0xe [decimal 14] pte contents:0x81 (valid 1, pfn 0x1 [decimal 1])

--&gt; Translates to Physical Address 0xb14 --&gt; Value: 07

Virtual Address 3350:

--&gt; pde index:0xc [decimal 12] pde contents:0xd4 (valid 1, pfn 0x54 [decimal 84])

--&gt; pte index:0x1a [decimal 26] pte contents:0xc0 (valid 1, pfn 0x40 [decimal 64])

--&gt; Translates to Physical Address 0x810 --&gt; Value: 04

Virtual Address 0a70:

--&gt; pde index:0x2 [decimal 2] pde contents:0xba (valid 1, pfn 0x3a [decimal 58])

--&gt; pte index:0x13 [decimal 19] pte contents:0x87 (valid 0, pfn 0x7 [decimal 7])

--&gt; Translates to Physical Address 0xf0 --&gt; Value: 1e

Virtual Address 55f9:

--&gt; pde index:0x15 [decimal 21] pde contents:0xd5 (valid 1, pfn 0x55 [decimal 85])

--&gt; pte index:0xf [decimal 15] pte contents:0x86 (valid 1, pfn 0x6 [decimal 6])

--&gt; Translates to Physical Address 0xd9 --&gt; Value: 1d

### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2~3 points.

### Full points:

40

### Score 1:

33.0

**Score 3:**

33

# Entry: 5

## Item: 6

### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run:

```
$ ./paging-multilevel-translate.py -s 3103
```

You are given the value of the PDBR, a complete dump of each page of memory, and a list of virtual addresses to translate.

Virtual Address 4a14: Translates to what Physical Address (and fetches what value)? Or Fault?

Virtual Address 4a14: --&gt; pde index:0x12 [decimal 18] pde contents:0x9a (valid 1, pfn 0x1a [decimal 26])

--&gt; pte index:0x10 [decimal 16] pte contents:0xd8 (valid 1, pfn 0x58 [decimal 88]) --&gt; Translates to Physical Address

Virtual Address 685e: Translates to what Physical Address (and fetches what value)? Or Fault?

Virtual Address 685e: --&gt; pde index:0x1a [decimal 26] pde contents:0xbf (valid 1, pfn 0x3f [decimal 63]) --&gt; pte index:

For each of the following virtual addresses, write down the physical address it translates to or write down that it is a fault (

Virtual Address 5a23

- Virtual Address 14ab
- Virtual Address 257e
- Virtual Address 7988
- Virtual Address 75cf
- Virtual Address 3350
- Virtual Address 0a70
- Virtual Address 55f9

### Answer:

Virtual Address 5a23:

--&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53])

--&gt; pte index:0x11 [decimal 17] pte contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

Virtual Address 14ab:

--&gt; pde index:0x5 [decimal 5] pde contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

Virtual Address 257e:

--&gt; pde index:0x9 [decimal 9] pde contents:0xc4 (valid 1, pfn 0x44 [decimal 68])

--&gt; pte index:0xb [decimal 11] pte contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

Virtual Address 7988:

--&gt; pde index:0x1e [decimal 30] pde contents:0xbb (valid 1, pfn 0x3b [decimal 59])

--&gt; pte index:0x8 [decimal 8] pte contents:0x19 (valid 0, pfn 0x19 [decimal 25])

--&gt; Fault (page table entry not valid)

Virtual Address 75cf:

--&gt; pde index:0x1d [decimal 29] pde contents:0xc6 (valid 1, pfn 0x46 [decimal 70])

--&gt; pte index:0xe [decimal 14] pte contents:0x81 (valid 1, pfn 0x1 [decimal 1])

--&gt; Translates to Physical Address 0xb14 --&gt; Value: 07

Virtual Address 3350:

--&gt; pde index:0xc [decimal 12] pde contents:0xd4 (valid 1, pfn 0x54 [decimal 84])

--&gt; pte index:0x1a [decimal 26] pte contents:0xc0 (valid 1, pfn 0x40 [decimal 64])

--&gt; Translates to Physical Address 0x810 --&gt; Value: 04

Virtual Address 0a70:

--&gt; pde index:0x2 [decimal 2] pde contents:0xba (valid 1, pfn 0x3a [decimal 58])

--&gt; pte index:0x13 [decimal 19] pte contents:0x87 (valid 0, pfn 0x7 [decimal 7])

--&gt; Translates to Physical Address 0xf0 --&gt; Value: 1e

Virtual Address 55f9:

--&gt; pde index:0x15 [decimal 21] pde contents:0xd5 (valid 1, pfn 0x55 [decimal 85])

--&gt; pte index:0xf [decimal 15] pte contents:0x86 (valid 1, pfn 0x6 [decimal 6])

--&gt; Translates to Physical Address 0xd9 --&gt; Value: 1d

### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte index:0x1 [decimal 1] pte contents:0x1a (valid 1, pfn 0x1a [decimal 26]) --&gt; Translates to Physical Address 0x1a5 --&gt; Value: 05

### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2~3 points.

### Full points:

40

### Score 1:

33



**Score 3:**

33

**Score outlier:**

21

## Entry: 6

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

5a23 - Fault (page table entry not valid) 14ab - Fault (page table entry not valid) 257e - Fault (page table entry not valid) 7

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

25.0

#### Score 3:

24

## Entry: 7

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

5a23 page table entry not valid. 14ab page directory entry not valid. 257e page table entry not valid. For 7988, the physic

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

30.0

#### Score 3:

30

## Entry: 8

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

40.0

#### Score 3:

40

## Entry: 9

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23 →pde index: 0x16 [decimal 22] pde contents: 0xb5 (valid 1, pfn 0x35 [decimal 53]) →pte index: 0x1

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

30.0

#### Score 3:

30

## Entry: 10

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: Fault (PTE not valid) Virtual Address 14ab: Fault (PDE not valid) Virtual Address 257e: Physical Ad

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

0.0

#### Score 3:

10

# Entry: 11

## Item: 6

### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

### Answer:

1. Virtual Address 5a23: pde index: 0x16 [decimal 22] pte index: 0x12 [decimal 18] pde contents: 0x3d (valid 1, pfn 0x1d p
2. Virtual Address 14ab: pde index: 0x05 [decimal 5] pte index: 0x15 [decimal 21] pde contents: 0x4f (valid 1, pfn 0x0f pte
3. Virtual Address 257e: pde index: 0x09 [decimal 9] pte index: 0x1f [decimal 31] pde contents: 0xb2 (valid 1, pfn 0x32 pt
4. Virtual Address 7988: pde index: 0x1e [decimal 30] pte index: 0x1c [decimal 28] pde contents: 0x2c (valid 1, pfn 0x0c p
5. Virtual Address 75cf: pde index: 0x1d [decimal 29] pte index: 0x0f [decimal 15] pde contents: 0x84 (valid 1, pfn 0x04 p
6. Virtual Address 3350: pde index: 0x0c [decimal 12] pte index: 0x14 [decimal 20] pde contents: 0x7e (valid 1, pfn 0x3e p
7. Virtual Address 0a70: pde index: 0x02 [decimal 2] pde contents: 0x6d (valid 1, pfn 0x2d [decimal 45]) Fault (page table
8. Virtual Address 55f9: pde index: 0x15 [decimal 21] pde contents: 0x8a (valid 1, pfn 0x0a [decimal 10]) Fault (page table

### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

### Full points:

40

### Score 1:

25.0

### Score 3:

10

## Entry: 12

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. For each of the following virtual addresses, write

#### Answer:

Virtual Address 5a23 - Fault  
Virtual Address 14ab - Fault  
Virtual Address 257e - Fault  
Virtual Address 7988 - Fault  
Virtual Address 75cf - Fault  
Virtual Address 3350 - Fault  
Virtual Address 0a70 - Fault  
Virtual Address 55f9 - Fault

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

10

#### Score 3:

15



## Entry: 13

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

5a23: Fault since 0x7f (valid bit = 0) 14ab: Fault since 0x7f (valid bit = 0) 257e: Fault, 7988: 0x19 75cf: 0x07, 3350: 0x04

#### Sample answer:

Virtual Address 5a23: --> pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --> pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

33.0

#### Score 3:

38

#### Score outlier:

25

## Entry: 14

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23 → Fault (page table entry not valid) Virtual Address 14ab → Fault (page directory entry not valid) Vi

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

25.0

#### Score 3:

17

## Entry: 15

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23 → Fault (page table entry not valid) Virtual Address 14ab → Fault (page directory entry not valid) Vi

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

25

#### Score 3:

17

## Entry: 16

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations.

#### Answer:

5a23: Page fault; 14ab: Page fault; 257e: Page fault; 7988: 0x1e; 75cf: 07; 3350: 0x04; 0a70: 0x1e; 55f9: 0x1d.

#### Sample answer:

Virtual Address 5a23: --> pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --> pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35

#### Score 3:

40

## Entry: 17

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

5a23 = fault ; 10110 10001 00011 = 22 17 3; 22th = b5 = 53; 17th = 7f = fault 14ab = 0b00001101 = 13 257e = fault 7988

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2~3 points.

#### Full points:

40

#### Score 1:

15.0

#### Score 3:

20

#### Score outlier:

40

## Entry: 18

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. For each of the following virtual addresses, write

#### Answer:

Virtual Address 5a23: Physical Address 53 (Valid) - Physical Address 127 (Fault)  
Virtual Address 14ab: Physical Address 127 (Fault) - Physical Address 127 (Fault)  
Virtual Address 257e: Physical Address 63 (Valid) - Physical Address 127 (Fault)  
Virtual Address 7988: Physical Address 59 (Valid) - Physical Address 29 (Valid)  
Virtual Address 75cf: Physical Address 70 (Valid) - Physical Address 1 (Valid)  
Virtual Address 3350: Physical Address 84 (Valid) - Physical Address 64 (Valid)  
Virtual Address 0a70: Physical Address 58 (Valid) - Physical Address 7 (Valid)  
Virtual Address 55f9: Physical Address 85 (Valid) - Physical Address 6 (Valid)

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35.0

#### Score 3:

40

# Entry: 19

## Item: 6

### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

### Answer:

Virtual Address 5a23: 101 1010 0010 0011 101 10=22 D5=0b1101 0101 101 0101= 85 10 001=17 7f 0111 1111 Fault  
Virtual Address 14ab: 0001 0010 1010 1011 0001 0 = 2 C2= 0b1100 0010 100 0010=0x42=66 10001=0x11=17 7f 0111 1  
Virtual Address 257e 0010 0101 0111 1110 0010 0=4 F8= 1111 1000 111 1000=0x78=120 1 0101=0x15=21 db 1101 10  
Virtual Address 7988 0111 1001 1000 1000 0111 1 =15 F9=1111 1001 111 1001=0x79=121 001 10=0x6=6 7f 0111 1111  
Virtual Address 75cf 0111 0101 1100 1111 0111 0 =15 F9=1111 1001 111 1001=0x79=121 101 11=0x17=23 7f 0111 11  
Virtual Address 3350 0011 0011 0101 0000 0011 0=6 7f 0111 1111 Fault  
Virtual Address 0a70 0000 1010 0111 000 0000 1 ff =1111 1111 111 1111=0x7f =127 010 01=0x9=9 7f 0111 1111 Fault  
Virtual Address 55f9 0101 0101 1111 1001 0101 0 = 10 C4=1100 0100 100 0100=0x44=68 101 11=0x17=23 7f 0111 11

### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2~3 points.

### Full points:

40

### Score 1:

23.0

### Score 3:

28

## Entry: 20

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103. F

#### Answer:

Virtual PDE Index PDE Contents PTE Index PTE Contents Physical Address Address

5a23 0x16 0xb5 (valid 1, pfn 0x11 [decimal 17] 0x7f (valid 0, pfn Fault (PTE [decimal 22] 0x35 [decimal 53]) 0x7f [decima

14ab 0x5 [decimal 0x7f (valid 0, pfn 0x7f Fault (PDE not Fault (PDE not valid) Fault (PDE 5) [decimal 127]) valid) not valid

257e 7988 75cf 3350 0a70 55f9

0x9 [decimal 0xc4 (valid 1, pfn 9] 0x44 [decimal 68])

0x1e 0xbb (valid 1, pfn [decimal 30] 0x3b [decimal 59])

0x1d 0xc6 (valid 1, pfn [decimal 29] 0x46 [decimal 70] 0xc [decimal 0xd4 (valid 1, pfn 12] 0x54 [decimal 84])

0x2 [decimal 0xba (valid 1, pfn 2] 0x3a [decimal 58])

0x15 0xd5 (valid 1, pfn [decimal 21] 0x55 [decimal 85])

0xb [decimal 11] 0x7f (valid 0, pfn 0x7f [decimal 127])

0xc [decimal 12] 0x9d (valid 1, pfn 0x1d [decimal 29])

0xe [decimal 14] 0x81 (valid 1, pfn 0x01 [decimal 1])

0x1a [decimal 26] 0xc0 (valid 1, pfn 0x40 [decimal 64])

0x13 [decimal 19] 0x87 (valid 1, pfn 0x07 [decimal 7] 0xf [decimal 15] 0x (valid 1, pfn [decimal 6])

Fault (PTE not valid)

Value: 19 (0x3a8)

Value: 07 (0x02f)

Value: 04 (0x810)

Value: 1e (0x0f0)

Value: 1d (0x0d9)

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35.0

#### Score 3:

40

#### Score outlier:

10



## Entry: 21

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: --&gt; Fault (page table entry not valid) Virtual Address 14ab: --&gt; Fault (page directory entry not

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2~3 points.

#### Full points:

40

#### Score 1:

35

#### Score 3:

40

## Entry: 22

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

5a23 (010110 10001 00011) pde index:0x16 [decimal 22] pde contents:0xd5 (valid 1, pfn 0x55 [decimal 85]) pte index:0x

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

25.0

#### Score 3:

15

## Entry: 23

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Answer for the 5a23: Pde index 22, content b5 --> valid 1 pfn = 53 Pte index 11, content 7f valid 0 (fault) It won't

#### Sample answer:

Virtual Address 5a23: --> pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --> pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

5.0

#### Score 3:

5

## Entry: 24

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Given virtual addresses, write down the physical

#### Answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35

#### Score 3:

40

## Entry: 25

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. For each of the following virtual addresses, write

#### Answer:

Virtual Address 5a23: Fault (page table entry not valid)

Virtual Address 14ab: Fault (page directory entry not valid)

Virtual Address 257e: Fault (page table entry not valid)

Virtual Address 7988: Translates to Physical Address 0x3a8

Virtual Address 75cf: Translates to Physical Address 0x2f

Virtual Address 3350: Translates to Physical Address 0x810

Virtual Address 0a70: Translates to Physical Address 0xf0

Virtual Address 55f9: Translates to Physical Address 0xd9

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35

#### Score 3:

40

## Entry: 26

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103

You are given the value of the PDBR, a complete dump of each page of memory, and a list of virtual addresses to translate.

Virtual Address 4a14:

--&gt; pde index:0x12 [decimal 18] pde contents:0x9a (valid 1, pfn 0x1a [decimal 26])

--&gt; pte index:0x10 [decimal 16] pte contents:0xd8 (valid 1, pfn 0x58 [decimal 88])

--&gt; Translates to Physical Address 0xb14 --&gt; Value: 17

Virtual Address 685e: Translates to what Physical Address (and fetches what value)? Or Fault?

Virtual Address 685e:

--&gt; pde index:0x1a [decimal 26] pde contents:0xbf (valid 1, pfn 0x3f [decimal 63])

--&gt; pte index:0x2 [decimal 2] pte contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

For each of the following virtual addresses, write down the physical address it translates to or write down that it is a fault (if it is).

Virtual Address 5a23

Virtual Address 14ab

Virtual Address 257e

Virtual Address 7988

Virtual Address 75cf

Virtual Address 3350

Virtual Address 0a70

Virtual Address 55f9

#### Answer:

5a23 --&gt; 10110 10001 00011, pde index=0b10110=22, content=0xb5=0b10110101, valid=1,PT=53

--&gt; pte index=0b10001 = 17, content=0x7f=0b01111111, valid=0, PT=127

--&gt; fault

14ab -&gt; 00101 00101 01011, pde index=5, content=0x7f, valid=0, PT=127

&gt; fault

257e -&gt; 01001 01011 11110, pde index=9, content=0xc4, valid=1, PT=68

&gt; pte index=11, content=0x7f, valid=0, PT=127

&gt; fault

7988 -&gt; 11110 01100 01000, pde index=30, content=0xbb, valid=1, PT=59

&gt; pte index=12, content=0x9d, valid=1, PT=29

&gt; offset=8, physical address=0b1110101000=0x3a8

75cf -&gt; 11101 01110 01111, index=29, content=0xc6, valid=1, PT=70

&gt; index=14, content=0x81, valid=1, PT=1

&gt; offset=15, physical address=0b0000101111=0x02f

3350 -&gt; 01100 11010 10000, index=12, content=0xd4, valid=1, PT=84

&gt; index=26, content=0xc0, valid=1, PT=64

&gt; offset=16, physical address=0b100000010000=0x810

0a70 -&gt; 00010 10011 10000, index=2, content=0xba, valid=1, PT=58

&gt; index=19, content=0x87, valid=1, PT=7

&gt; offset=16, physical address=0b0011110000=0x0f0

55f9 -&gt; 10101 01111 11001, index=21, content=0xd5, valid=1, PT=85

&gt; index=15, content=0x86, valid=1, PT=6

&gt; offset=25, physical address=0b0011011001=0x0d9

### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2~3 points.

### Full points:

40

### Score 1:

30.0

**Score 3:**

35



## Entry: 27

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

1. Fault 2. Fault 3. Fault 4. 25 5. 7 6. 4 7. 30 8. 29 7 CS3103 - Operating Systems

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35

#### Score 3:

40

## Entry: 28

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

VA 0x5a23: PDE index 0b10110=22, PDE content 0xb5=0b10110101, valid, PT=0b0110101=0x35=53 PTE index 0b100

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2~3 points.

#### Full points:

40

#### Score 1:

30.0

#### Score 3:

35

## Entry: 29

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

5a23: page table entry not valid. 14ab: page directory entry not valid 257e: page table entry not valid. 7988: page table en

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

25.0

#### Score 3:

27

## Entry: 30

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: Fault • Virtual Address 14ab: Fault • Virtual Address 257e: 0xb14 • Virtual Address 7988: 0x3a8 • V

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

30.0

#### Score 3:

35

# Entry: 31

## Item: 6

### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

### Answer:

5a23: Fault

14ab: Fault

257e: Fault

7988: Physical address: 0x3a8, Value: 19

75cf: Physical address: 0x2f, Value: 07

3350: Physical address: 0x810, Value: 04

0a70: Physical address: 0xf0, Value: 1e

55f9: Physical address: 0xd9, Value: 1d

### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

### Full points:

40

### Score 1:

35

### Score 3:

40

## Entry: 32

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35

#### Score 3:

40

## Entry: 33

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-tr...

#### Answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

30.0

#### Score 3:

40

# Entry: 34

## Item: 6

### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run:  
\$ ./paging-multilevel-translate.py -s 3103

For each of the following virtual addresses, write down the physical address it translates to or write down that it is a fault (

Virtual Address 5a23

Virtual Address 14ab

Virtual Address 257e

Virtual Address 7988

Virtual Address 75cf

Virtual Address 3350

Virtual Address 0a70

Virtual Address 55f9

### Answer:

Virtual Address 5a23:

--&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 51])

--&gt; pte index:0x11 [decimal 17] pte contents:0x00 (valid 0, pfn 0x00 [decimal 0])

--&gt; Fault (page table entry not valid)

Virtual Address 14ab

--&gt; pde index:0x05 [decimal 5] pde contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page directory entry not valid)

Virtual Address 257e

--&gt; pde index:0x09 [decimal 9] pde contents:0xc4 (valid 1, pfn 0x44 [decimal 68])

--&gt; pte index:0x0b [decimal 11] pte contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

Virtual Address 7988

--&gt; pde index:0x1e [decimal 30] pde contents:0xbb (valid 1, pfn 0x3b [decimal 59])

--&gt; pte index:0x0c [decimal 12] pte contents:0x9d (valid 1, pfn 0x1d [decimal 29])

--&gt; Translates to Physical Address 0x528

Virtual Address 75cf

--&gt; pde index:0x1d [decimal 29] pde contents:0xc6 (valid 1, pfn 0x46 [decimal 70])

--&gt; pte index:0x06 [decimal 6] pte contents:0x7f (valid 0, pfn 0x7f [decimal 127])

--&gt; Fault (page table entry not valid)

Virtual Address 3350

--&gt; pde index:0x0c [decimal 12] pde contents:0xd4 (valid 1, pfn 0x54 [decimal 84])



--&gt; pte index:0x1a [decimal 26] pte contents:0xc0 (valid 1, pfn 0x40 [decimal 64])  
--&gt; Translates to Physical Address 0xc90

Virtual Address 0a70

--&gt; pde index:0x02 [decimal 2] pde contents:0xba (valid 1, pfn 0x3a [decimal 58])  
--&gt; pte index:0x13 [decimal 19] pte contents:0x87 (valid 1, pfn 0x07 [decimal 7])  
--&gt; Translates to Physical Address 0xf0

Virtual Address 55f9

--&gt; pde index:0x15 [decimal 21] pde contents:0xd5 (valid 1, pfn 0x55 [decimal 85])  
--&gt; pte index:0x0f [decimal 15] pte contents:0x86 (valid 1, pfn 0x06 [decimal 6])  
--&gt; Translates to Physical Address 0xd9

### **Sample answer:**

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte index:0x0a [decimal 10] pte contents:0x8b (valid 1, pfn 0x0b [decimal 3]) --&gt; Translates to Physical Address 0x35b

### **Sample criteria:**

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2~3 points.

### **Full points:**

40

### **Score 1:**

25.0

### **Score 3:**

24

## Entry: 35

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: physical address: 0x07023: 7  
Virtual Address 14ab: physical address: 0x0c04b: 13  
Virtual Address 257e: physical address: 0x0007e: 0  
Virtual Address 7988: fault  
Virtual Address 75cf: fault  
Virtual Address 3350: physical address: 0x00050: 0  
Virtual Address 0a70: fault  
Virtual Address 55f9: physical address: 0x0bf9: 11

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

0.0

#### Score 3:

0

# Entry: 36

## Item: 6

### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

Virtual Address 4a14: Translates to what Physical Address (and fetches what value)? Or Fault?

Virtual Address 4a14: --&gt; pde index:0x12 [decimal 18] pde contents:0x9a (valid 1, pfn 0x1a [decimal 26])  
--&gt; pte index:0x10 [decimal 16] pte contents:0xd8 (valid 1, pfn 0x58 [decimal 88])  
--&gt; Translates to Physical Address 0xb14  
--&gt; Value: 17

Virtual Address 685e: Translates to what Physical Address (and fetches what value)? Or Fault?

Virtual Address 685e: --&gt; pde index:0x1a [decimal 26] pde contents:0xbf (valid 1, pfn 0x3f [decimal 63])  
--&gt; pte index:0x2 [decimal 2] pte contents:0x7f (valid 0, pfn 0x7f [decimal 127])  
--&gt; Fault (page table entry not valid)

For each of the following virtual addresses, write down the physical address it translates to or write down that it is a fault (

Virtual Address 5a23

- Virtual Address 14ab
- Virtual Address 257e
- Virtual Address 7988
- Virtual Address 75cf
- Virtual Address 3350
- Virtual Address 0a70
- Virtual Address 55f9

### Answer:

Virtual Address 5a23, physical address: 0xfe3, value: 7f  
Virtual Address 14ab, physical address: 0xfeb, value: 7f  
Virtual Address 257e, physical address: 0xffe, value: 7f  
Virtual Address 7988, physical address: 0x3a8, value: 19  
Virtual Address 75cf, physical address: 0x0ef, value: 0f  
Virtual Address 3350, physical address: 0x810, value: 04  
Virtual Address 0a70, physical address: 0x0f0, value: 1e  
Virtual Address 55f9, physical address: 0x0d9, value: 18

### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

### Full points:

40

**Score 1:**

0.0

**Score 3:**

0

## Entry: 37

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: PDE index: 0x16 [decimal 22] PDE contents: 0x7F (valid 0, pfn 0x7f [decimal 127]) Fault (page dict

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35

#### Score 3:

40

## Entry: 38

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23 Fault (page table entry not valid)  
Virtual Address 14ab Fault page directory entry not valid  
Virtual Address 257e Fault (page table entry not valid)  
Virtual Address 7988 Value :25  
Virtual Address 75cf Fault (page table entry not valid)  
Virtual Address 3350 Value:127  
Virtual Address 0a70 Fault (page table entry not valid)  
Virtual Address 55f9 Value:29

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

15.0

#### Score 3:

20

## Entry: 39

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Run: \$ ./paging-multilevel-translate.py -s 3103 Y

#### Answer:

Virtual Address 5a23: Fault Virtual Address 14ab: Physical Address 0d Virtual Address 257e: Fault Virtual Address 7988:

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

15.0

#### Score 3:

20

## Entry: 40

### Item: 6

#### Question:

Use the simulator paging-multilevel-translate.py to perform translations. Virtual Address 5a23, 14ab, 257e, 7988, 75cf, 33

#### Answer:

5a23: pde index:0x16 [22] pde contents:0xb5 (valid 1, pfn 0x35 [53]) pte index:0x11 [decimal 17] pte contents:0x7f (valid

#### Sample answer:

Virtual Address 5a23: --&gt; pde index:0x16 [decimal 22] pde contents:0xb5 (valid 1, pfn 0x35 [decimal 53]) --&gt; pte ind

#### Sample criteria:

7 virtual address, each address for 5 points

- For each address, if the answer “final physical address it translates to” or “write down that it is a fault” correctly (7 points)
- If the answer is partially correct but has detailed procedures, can give 2–3 points.

#### Full points:

40

#### Score 1:

35.0

#### Score 3:

40